Straight Line

Contents

1. Coordinates

- Cartesian Coordinates (x, y) to Polar Coordinates (r, θ):

$$r = \sqrt{x^2 + y^2}$$

$$\begin{aligned} \theta &= \tan^{-1}\left(\frac{y}{x}\right) \quad \text{in Quadrant I} \\ &= \pi - \tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in Quadrant II} \\ &= \pi + \tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in Quadrant III} \\ &= -\tan^{-1}\left(\left|\frac{y}{x}\right|\right) \quad \text{in Quadrant IV} \end{aligned}$$

- Polar Coordinates (r, θ) to Cartesian Coordinates (x, y):

$$x = r\cos\theta$$

$$y = r\sin\theta$$

- Distance between two points in Cartesian Coordinates $P(x_1, y_1)$ and $Q(x_2, y_2)$:

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

- Distance between two points in Polar Coordinates $P(r_1, \theta_1)$ and $Q(r_2, \theta_2)$:

$$d = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos(\theta_1 - \theta_2)}$$

- Section Formula (Internal Division):

$$x = \frac{m_1 x_2 + m_2 x_1}{m_1 + m_2}, \quad y = \frac{m_1 y_2 + m_2 y_1}{m_1 + m_2}$$

- Section Formula (External Division):

$$x = \frac{m_1 x_2 - m_2 x_1}{m_1 - m_2}, \quad y = \frac{m_1 y_2 - m_2 y_1}{m_1 - m_2}$$

- Midpoint Formula:

$$M = \left(\frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2}\right)$$

- Area of Triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$\Delta ABC = \frac{1}{2}\begin{vmatrix} x_1 & y_1 & 1 \\ x_2 & y_2 & 1 \\ x_3 & y_3 & 1 \end{vmatrix}$$

- Centroid G of Triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$:

$$G = \left(\frac{x_1 + x_2 + x_3}{3}, \frac{y_1 + y_2 + y_3}{3}\right)$$

- Incenter P of Triangle ABC with vertices $(x_1, y_1), (x_2, y_2), (x_3, y_3)$ and a, b, c side lengths:

$$P = \left(\frac{ax_1 + bx_2 + cx_3}{a + b + c}, \frac{ay_1 + by_2 + cy_3}{a + b + c}\right)$$

- Excenter E_A opposite to vertex A:

$$E_A = \left(\frac{-ax_1 + bx_2 + cx_3}{-a + b + c}, \frac{-ay_1 + by_2 + cy_3}{-a + b + c}\right)$$

- Circumcenter O of Triangle ABC:

$$O = \left(\frac{x_1 \sin 2A + x_2 \sin 2B + x_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}, \frac{y_1 \sin 2A + y_2 \sin 2B + y_3 \sin 2C}{\sin 2A + \sin 2B + \sin 2C}\right)$$

- Orthocenter H of Triangle ABC:

$$H = \left(\frac{x_1 \tan A + x_2 \tan B + x_3 \tan C}{\tan A + \tan B + \tan C}, \frac{y_1 \tan A + y_2 \tan B + y_3 \tan C}{\tan A + \tan B + \tan C}\right)$$

2. Straight Lines

- General Form of a Straight Line:

$$ax + by + c = 0$$

- Slope (m) of a line passing through $(x_1, y_1), (x_2, y_2)$, making an angle θ with the positive x-axis:

$$m = \tan\theta = \frac{y_2 - y_1}{x_2 - x_1}$$

- Slope-Intercept Form (Slope m, y-intercept c):

$$y = mx + c$$

- Point-Slope Form (Passes through (x_1, y_1) with slope m):

$$y - y_1 = m(x - x_1)$$

- Two-Point Form (Passes through (x_1, y_1) and (x_2, y_2)):

$$\frac{y - y_1}{y_2 - y_1} = \frac{x - x_1}{x_2 - x_1}$$

- Intercept Form (x-intercept a, y-intercept b):

$$\frac{x}{a} + \frac{y}{b} = 1$$

- Normal (Perpendicular) Form (Perpendicular distance from origin p, angle α with positive x-axis):

$$x\cos\alpha + y\sin\alpha = p$$

- Parametric / Symmetric Form (Distance r from (x_1, y_1) along angle θ):

$$\frac{x - x_1}{\cos\theta} = \frac{y - y_1}{\sin\theta} = r$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be coincident:

$$\frac{a_1}{a_2} = \frac{b_1}{b_2} = \frac{c_1}{c_2}$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be parallel to each other:

$$m_1 = m_2 \Rightarrow \frac{a_1}{a_2} = \frac{b_1}{b_2} \neq \frac{c_1}{c_2}$$

- The equation of a line parallel to $ax + by + c = 0$:

$$ax + by + k = 0$$

- Conditions for $a_1x + b_1y + c_1 = 0$ and $a_2x + b_2y + c_2 = 0$ to be perpendicular to each other:

$$m_1m_2 = -1 \Rightarrow a_1a_2 + b_1b_2 = 0$$

- The equation of the line orthogonal to $ax + by + c = 0$:

$$bx - ay + k = 0$$

- The angle between two lines with slopes m_1 and m_2:

$$\tan\theta = \pm\frac{m_1 - m_2}{1 + m_1 m_2}$$

- Perpendicular distance from (x_1, y_1) to $ax + by + c = 0$:

$$d = \frac{|ax_1 + by_1 + c|}{\sqrt{a^2 + b^2}}$$

- Perpendicular distance between two parallel lines $ax + by + c_1 = 0$, $ax + by + c_2 = 0$:

$$d = \frac{|c_1 - c_2|}{\sqrt{a^2 + b^2}}$$

- Foot of Perpendicular (h, k) from (x_1, y_1) to $ax + by + c = 0$:

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -\frac{ax_1 + by_1 + c}{a^2 + b^2}$$

- Image / Reflection (h, k) of (x_1, y_1) in line $ax + by + c = 0$:

$$\frac{h - x_1}{a} = \frac{k - y_1}{b} = -2\frac{ax_1 + by_1 + c}{a^2 + b^2}$$

- Equation of a line that bisects the angle between two intersecting lines $(a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0)$:

$$\frac{a_1x + b_1y + c_1}{\sqrt{a_1^2 + b_1^2}} = \pm\frac{a_2x + b_2y + c_2}{\sqrt{a_2^2 + b_2^2}}$$

  1. If $a_1a_2 + b_1b_2 > 0$:
     - The + sign gives you obtuse angle bisector.
     - The - sign gives you acute angle bisector.
  2. If $a_1a_2 + b_1b_2 < 0$:
     - The + sign gives you acute angle bisector.
     - The - sign gives you obtuse angle bisector.

- Family of lines passing through the intersection of $L_1 = 0$ and $L_2 = 0$:

$$L_1 + \lambda L_2 = 0$$

- Conditions for concurrency of three lines ($a_1x + b_1y + c_1 = 0, a_2x + b_2y + c_2 = 0, a_3x + b_3y + c_3 = 0$):

$$\begin{vmatrix} a_1 & b_1 & c_1 \\ a_2 & b_2 & c_2 \\ a_3 & b_3 & c_3 \end{vmatrix} = 0$$

- Pair of straight lines passing through origin ($ax^2 + 2hxy + by^2 = 0$):

$$\tan\theta = \frac{2\sqrt{h^2 - ab}}{|a + b|}$$